Fundamental Theorem of Galois Theory

An End-to-End Meta-Math Walkthrough

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Abstract

A complete authored walkthrough of fundamental theorem of galois theory, with explicit mathematical contracts, local verification gates, and reusable exports. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

MA-04 / Modern Algebra / Report v1

I. TARGET AND IMPORT CONTRACT

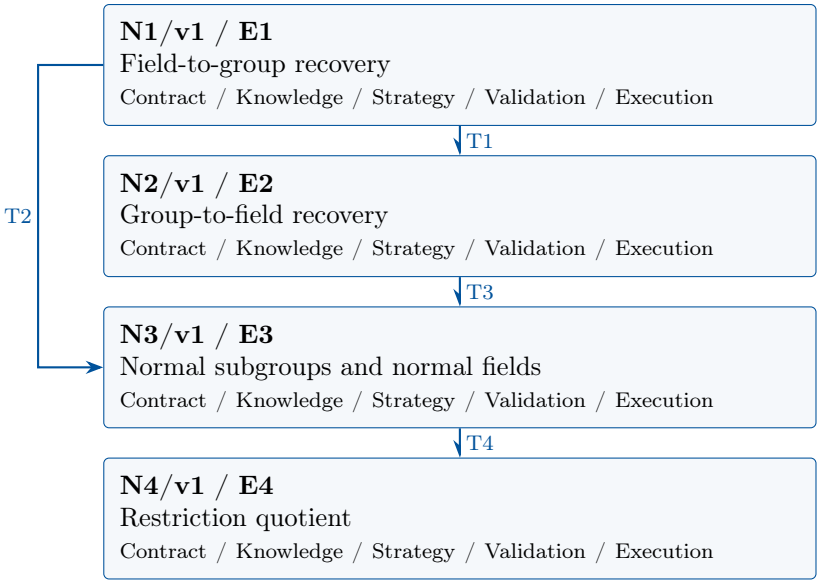
Theorem I.1. *For a finite Galois extension L/K with group G, prove an inclusion-reversing bijection between intermediate fields E and subgroups $H \leq G$, via $E \mapsto \mathrm{Gal}(L/E)$ and $H \mapsto L^H$. Prove $[L : L^H] = |H|$ and normal subgroups correspond to intermediate extensions Galois over K, with quotient group G/H.*

A. Foundations and permitted imports

L/K is a finite normal separable extension, $G = \mathrm{Aut}_K(L)$. Import: (i) finite Galois extensions have automorphism count equal to degree; (ii) Artin's fixed-field theorem, for a finite automorphism subgroup H, $[L : L^H] = |H|$ and L/L^H is Galois; (iii) the tower law; (iv) embeddings of an intermediate field into an algebraic closure extend to L, and K-normality sendsL to itself. These are substantive field-theory imports, not the correspondence being proved.

The last step also imports the first isomorphism theorem (MA-02).

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1: N1/E1/v1 to N2. The field recovered from its fixing subgroup is exactlyE.

Receiving activity. N2: Group-to-field recovery

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: SubgroupH $\leq G$; $E_H = L^H$; prior exports E1/v1

T2: N1/E1/v1 to N3. The field recovered from its fixing subgroup is exactlyE.

Receiving activity. N3: Normal subgroups and normal fields

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Correspondence and extension of embeddings; prior exports E1/v1, E2/v1

T3: N2/E2/v1 to N3. Both inverse identities and the degree $formula[L : L^H] = |H|$.

Receiving activity. N3: Normal subgroups and normal fields

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Correspondence and extension of embeddings; prior exports E1/v1, E2/v1

T4: N3/E3/v1 to N4. H normal in G iff its fixed field is Galois over K.

Receiving activity. N4: Restriction quotient

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: NormalH and $E = L^H$; prior exports E3/v1

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Field-to-group recovery

Object and role. Compare degrees after forming its fixing subgroup.

Inputs. Intermediate $K \subset E \subset L$; foundational imports stated in §1

C — Contract

Own E1/v1: The field recovered from its fixing subgroup is exactlyE.

K — Knowledge

Retain Intermediate $K \subset E \subset L$; foundational imports stated in §1; keep their assumptions and versions attached.

S — Strategy

Compare degrees after forming its fixing subgroup.

V — Validation

Inclusion alone is not enough; degree equality closes recovery.

E — Execution

Carry out the numbered deductions and publish E1/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N1.1 L/E remains normal and separable: L is a splitting field over K of separable polynomials and hence also a splitting field of those polynomials over E.

N1.2 Set $H_E = \mathrm{Aut}_E(L)$; the automorphism-count import $gives |H_E| = [L : E]$.

N1.3 Certainly $E \subset L^{H_E}$.

N1.4 Artin $gives [L : L^{H_E}] = |H_E| = [L : E]$.

N1.5 The tower law then $forces [L^{H_E} : E] = 1$, so $L^{H_E} = E$.

Export E1/v1. The field recovered from its fixing subgroup is exactlyE.

Acceptance gate. Inclusion alone is not enough; degree equality closes recovery. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N1: Inclusion alone is not enough; degree equality closes recovery. Recheck dependent nodes N2, N3, N4.

B. N2/v1: Group-to-field recovery

Object and role. Compare the group sizes supplied by Artin.

Inputs. SubgroupH $\leq G$; $E_H = L^H$; prior exports E1/v1

C — Contract

Own E2/v1: Both inverse identities and the degree $formula[L : L^H] = |H|$.

K — Knowledge

Retain SubgroupH $\leq G$; $E_H = L^H$; prior exports E1/v1; keep their assumptions and versions attached.

S — Strategy

Compare the group sizes supplied by Artin.

V — Validation

Fixing all of E is stronger than preservingE as a set.

E — Execution

Carry out the numbered deductions and publish E2/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N2.1 H fixes E_H pointwise, so $H \subset \mathrm{Aut}_{E_H}(L)$.

N2.2 Artin says L/E_H is Galois and its degree is|H|.

N2.3 Thus its full automorphism group has|H| elements and equals H.

N2.4 The two constructions are therefore mutual inverses.

N2.5 If $E_1 \subset E_2$ then every automorphism fixing E_2 fixes E_1, so $H_{E_2} \subset H_{E_1}$; if $H_1 \subset H_2$ then $L^{H_2} \subset L^{H_1}$.

N2.6 This proves the inclusion-reversing bijection.

Export E2/v1. Both inverse identities and the degree $formula[L : L^H] = |H|$.

Acceptance gate. Fixing all of E is stronger than preservingE as a set. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N2: Fixing all of E is stronger than preservingE as a set. Recheck dependent nodes N3, N4.

C. N3/v1: Normal subgroups and normal fields

Object and role. Translate conjugation to stability of the fixed field.

Inputs. Correspondence and extension of embeddings; prior exports E1/v1, E2/v1

C — Contract

Own E3/v1: H normal in G iff its fixed field is Galois over K.

K — Knowledge

Retain Correspondence and extension of embeddings; prior exports E1/v1, E2/v1; keep their assumptions and versions attached.

S — Strategy

Translate conjugation to stability of the fixed field.

V — Validation

All embeddings, not just a few automorphisms, must be controlled.

E — Execution

Carry out the numbered deductions and publish E3/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N3.1 For $\sigma \in G$, the fixing subgroup of $\sigma(E)$ is $\sigma H_E \sigma^{-1}$, so the correspondence makes H_E normal exactly when $\sigma(E) = E$ for every σ.

N3.2 If E/K is normal, each such σ preserves E.

N3.3 Conversely every K-embedding of E extends to aK-embedding of L and thus to an element of G; stability then makes all conjugates of E equal E, proving normality.

N3.4 Separability descends from L/K, so this is precisely the Galois condition on E/K.

Export E3/v1. H normal in G iff its fixed field is Galois over K.

Acceptance gate. All embeddings, not just a few automorphisms, must be controlled. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N3: All embeddings, not just a few automorphisms, must be controlled. Recheck dependent nodes N4.

D. N4/v1: Restriction quotient

Object and role. Check surjectivity using embedding extension.

Inputs. NormalH and $E = L^H$; prior exports E3/v1

C — Contract

Own E4/v1: The normal-field quotient identification $G/H \cong \mathrm{Gal}(E/K)$.

K — Knowledge

Retain NormalH and $E = L^H$; prior exports E3/v1; keep their assumptions and versions attached.

S — Strategy

Check surjectivity using embedding extension.

V — Validation

Without normality restriction need not land in $\mathrm{Aut}_K(E)$.

E — Execution

Carry out the numbered deductions and publish E4/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N4.1 Restriction res : $G \to \mathrm{Aut}_K(E)$ is defined because E is stable.

N4.2 Its kernel consists exactly of automorphisms fixing E, namelyH.

N4.3 AnyK-automorphism of E extends to L; normality of L/K makes the extension an element of G, so res is onto.

N4.4 By MA-02, $G/H \cong \mathrm{Aut}_K(E) = \mathrm{Gal}(E/K)$.

N4.5 This closes the entire theorem.

Export E4/v1. The normal-field quotient identification $G/H \cong \mathrm{Gal}(E/K)$.

Acceptance gate. Without normality restriction need not land in $\mathrm{Aut}_K(E)$. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N4: Without normality restriction need not land in $\mathrm{Aut}_K(E)$. Withhold this terminal result until its gate passes.

IV. CONTINUOUS MATHEMATICAL PROOF

Read these steps in order as one proof; the diagram is optional. Numbered activities identify the specific constructions and deductions owned by each node.

A. §4.N1 — Field-to-group recovery

1. L/E remains normal and separable: L is a splitting field over K of separable polynomials and hence also a splitting field of those polynomials over E.
2. Set $H_E = \mathrm{Aut}_E(L)$; the automorphism-count import $gives |H_E| = [L : E]$.
3. Certainly $E \subset L^{H_E}$.
4. Artin $gives [L : L^{H_E}] = |H_E| = [L : E]$.
5. The tower law then $forces [L^{H_E} : E] = 1$, so $L^{H_E} = E$.

B. §4.N2 — Group-to-field recovery

1. H fixes E_H pointwise, so $H \subset \mathrm{Aut}_{E_H}(L)$.
2. Artin says L/E_H is Galois and its degree is|H|.
3. Thus its full automorphism group has|H| elements and equals H.
4. The two constructions are therefore mutual inverses.
5. If $E_1 \subset E_2$ then every automorphism fixing E_2 fixes E_1, so $H_{E_2} \subset H_{E_1}$; if $H_1 \subset H_2$ then $L^{H_2} \subset L^{H_1}$.
6. This proves the inclusion-reversing bijection.

C. §4.N3 — Normal subgroups and normal fields

1. For $\sigma \in G$, the fixing subgroup of $\sigma(E)$ is $\sigma H_E \sigma^{-1}$, so the correspondence makes H_E normal exactly when $\sigma(E) = E$ for every σ.
2. If E/K is normal, each such σ preserves E.
3. Conversely every K-embedding of E extends to aK-embedding of L and thus to an element of G; stability then makes all conjugates of E equal E, proving normality.
4. Separability descends from L/K, so this is precisely the Galois condition on E/K.

D. §4.N4 — Restriction quotient

1. Restriction res : $G \to \mathrm{Aut}_K(E)$ is defined because E is stable.
2. Its kernel consists exactly of automorphisms fixing E, namelyH.
3. AnyK-automorphism of E extends to L; normality of L/K makes the extension an element of G, so res is onto.
4. By MA-02, $G/H \cong \mathrm{Aut}_K(E) = \mathrm{Gal}(E/K)$.
5. This closes the entire theorem.

□

V. WORKED ORCHESTRATION

Let $L = Q(\sqrt{2}, \sqrt{3})$, $K = Q$. The four sign changes form $C_2 \times C_2$. The subgroup fixing $\sqrt{2}$ consists of the identity and the change $\sqrt{3 \mapsto} - \sqrt{3}$; its fixed field is $Q(\sqrt{2})$, $and[L : Q(\sqrt{2})] = 2 = |H|$. All subgroups are normal and the restriction quotient has order 2.

A. Boundary and hypothesis audit

$Q(\sqrt[3]{2})/Q$ is not normal: its nonreal conjugate roots are absent and itsQ-automorphism group is trivial. Its fixed field under that trivial group is the whole extension, notQ. Validation blocks using this extension in the finite Galois correspondence.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Failure and correction

This is an authored teaching trace, not a recorded agent run.

1. **Contract → Knowledge:** register the §1 theorem, imports, and requested exports before beginning the proof.
2. **Strategy → Execution:** choose the bridges in §3 and carry out the numbered activities in §4; transmit only established E-exports along the §2 DAG.
3. **Validation → responsible node:** test the specific boundary/failure in §5 and the node's individual audit. A failing candidate is withheld; downstream nodes may not use it.

4. **Knowledge + Strategy:** identify whether the error is an omitted condition, an invalid inference, or an attempted extension beyond the theorem. Restore the exact contract or explicitly state a different one.
5. **Execution:** redo the affected mathematical activity using the repaired condition or inference. Recheck consuming nodes, rather than feeding their conclusions back as premises.
6. **Contract:** publish only the supported export with its assumptions and proof location. A hypothetical revised export is v2; the written proof here is the v1 exposition, not a claim that this repair was executed.

B. Hypothesis boundary

$Q(\sqrt[3]{2})/Q$ is not normal: its nonreal conjugate roots are absent and itsQ-automorphism group is trivial. Its fixed field under that trivial group is the whole extension, notQ. Validation blocks using this extension in the finite Galois correspondence.

C. Version and recheck

Treat any changed assumption as a new contract. Recheck the revised export and all of its descendants. A backward diagnostic never establishes an earlier theorem. This is an authored teaching trace, not an execution log.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading.

- MIT Algebra II notes: field extensions and Galois theory.